Typed Decision Models as Passage Rerankers: Effectiveness, Calibration, and Context Selection

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Abstract

Typed decision models are language models that return a probability for each option in a caller-defined answer set. We test four such model families, without task-specific tuning, as pointwise rerankers of the top 100 BM25 candidates on 13 collections from TREC Deep Learning and BEIR. The closed API model Jev averages 55.3–56.1 nDCG@10 (normalized discounted cumulative gain at rank 10) across three question formats, on par with a 4-bit Qwen3-Reranker-8B (56.0) and above the cross-encoder bge-reranker-v2-m3 (53.7), with wide variation across collections. In same-base controls, Clef differs significantly from its base language model on one of seven collections, while Jeff improves on its base. Jev's raw yes/no probabilities have lower expected calibration error than bge's (15.0 vs. 20.0), but bge calibrated on collections with the same relevance threshold closes this gap, and rewording the prompt moves raw calibration error by more than the gap. In retrieval-augmented generation on a family-disjoint split of MuSiQue, Jev raises answer F1 over bge in all 27 policy–generator pairs (23 significant in family-clustered tests) and over Qwen3-Reranker-8B in 26 (12 significant); on Natural Questions, one of 27 differences is significant. Probability thresholds do not beat fixed top-k selection. Typed decision models are competitive rerankers, but calibration alone does not improve context selection.

Keywords: reranking, decision models, calibration, retrieval-augmented generation, BEIR

1 Introduction

Retrieval-augmented generation (RAG) pipelines typically retrieve candidates with a cheap first stage, rerank them with a cross-encoder [41] or an LLM [57], and pass a fixed top-k to the generator. Two weaknesses of this recipe are well known: cross-encoders trained on MS MARCO transfer imperfectly to new domains [60], and their scores are not probabilities [43], so the cut-off k is chosen by habit rather than by evidence.

A recently released class of models targets both: *typed decision models* such as Jev [64], Cloudflare Clef [8], Jeff [22], and Laya [11]. Instead of generating text, they answer a typed question about a state—a choice among named options, a yes/no (Noul) question, or an ordered scale—and return per-option probabilities. A relevance judgment is a natural typed question (*Is this passage relevant to the query?*), and a per-option probability is exactly what a context-selection rule such as "keep passages with $P(\text{relevant}) > t$" needs. Whether that probability is actually calibrated, and whether such a threshold beats a fixed cut-off, are empirical questions that the interface alone does not answer.

Evidence so far is thin: decision models have been evaluated for recommendation reranking [34] and agent-memory selection [56], a practitioner blog reports Jev results on two BEIR datasets [27], and an early audit finds that typed readouts have not yet shown an accuracy advantage over comparable label-probability readouts [59]. No systematic study has evaluated decision models as passage rerankers on standard IR benchmarks against cross-encoders and label-probability LLM rerankers, or measured whether their relevance probabilities are calibrated.

We provide that study, organized around three research questions. **RQ1:** Can typed decision models, used without task-specific tuning as pointwise rerankers, match cross-encoders and label-probability LLM rerankers? **RQ2:** Are their relevance probabilities calibrated, with and without labels from the target collection? **RQ3:** Do these probabilities help decide which passages a retrieval-augmented generator should see? Our contributions and main findings are: (1) An evaluation of four decision-model families and three question formats on 13 collections against cross-encoders and label-probability LLM rerankers, with same-base controls for Clef and Jeff and equal-context ablations. The large decision models are on par on average with the strongest label-probability reranker we run on all 13 collections, a 4-bit Qwen3-Reranker-8B, with large differences on individual collections; Clef is close to its own base LLM, and Jev's base is unknown. (2) A calibration analysis on nine collections with judged negatives. Raw yes/no probabilities from Jev and Laya are better calibrated than raw cross-encoder scores, but a cross-encoder calibrated on other collections with the same relevance threshold closes the gap, so the practical advantage is not needing labelled source data; raw calibration also depends strongly on the prompt wording. (3) A RAG evaluation on MuSiQue and NQ with three generators, comparing fixed top-k, gap-based, and threshold-based context selection. Jev's rankings help multi-hop QA more than those of bge-reranker-v2-m3 or the equally effective Qwen3-Reranker-8B, including on a family-disjoint split, and Jev more often ranks the answer-bearing hop first, yet probability thresholds do not beat fixed k. (4) Code for reproducing and extending the evaluation, with input hashes, split IDs, and all statistical outputs.

2 Related Work

2.1 Neural and LLM reranking

Cross-encoders score each query–passage pair jointly [41, 42] and remain the default second stage; BEIR showed that their in-domain gains shrink out of domain [60]. LLM rerankers can be pointwise, pairwise, setwise, or listwise: RankGPT's sliding-window listwise prompting is competitive with supervised rerankers [57], distilled listwise models such as RankZephyr and FIRST make this cheaper [45, 50], and setwise prompting trades off the two [73]. Pointwise LLM rankers read relevance from the probability of a "yes" token and improve with graded rather than binary labels [72]; Qwen3-Reranker is a recent model trained for exactly this readout [71]. Typed decision models are also pointwise, but their answer space is a declared set of options, each of which receives a probability by construction; we compare them directly with label-probability rerankers.

2.2 LLMs as relevance assessors

LLMs can predict human relevance preferences [61] and are used to generate relevance labels at scale [65], with ongoing debate about the risks of doing so [10, 21]; the LLMJudge challenge [48] benchmarks such assessors on TREC DL. These studies measure label agreement; we measure whether the probabilities are calibrated.

2.3 Typed decision models

Jev [64], its open counterparts Jeff [22] and Laya [11], and Cloudflare's Clef [8] share a request format of a state plus typed questions. The public JevBench leaderboard [36] (not peer reviewed) ranks such models on accuracy, calibration, speed, and cost, but reports no retrieval tasks. The closest academic work evaluates Jev for product recommendation reranking [34]; others use it to select agent-memory records [56]. A study of the fine-tuned Laya typed-decisions checkpoint finds it uniformly under-confident on workflow decisions [38], and Jev is reported calibrated on familiar closed-choice tasks but over-confident beyond its knowledge boundary [54]; an early audit of 28 papers on Jev finds its clearest gains in latency and cost rather than accuracy over comparable label-probability readouts [59], which is the comparison our label-probability baselines make for retrieval. Concurrent work concludes that System One decision models are not better rerankers [20]. In non-peer-reviewed reports, practitioners have described Jev reranking results on SciFact and NFCorpus [27] and found Jev below monoBERT when reranking the roughly 1,000 MS MARCO candidates per DL19 query [6]; we rerank only the BM25 top-100.

2.4 Calibrated relevance and context selection

Neural rankers are known to be poorly calibrated [43], as are modern neural networks generally [25], and ranking losses leave the scale of scores unconstrained, which motivates scale-calibrated training [69]; Platt scaling [44] is the standard fix but needs labels. LLM rerankers' verbalized confidence in their own ranking quality is severely

Table 1 The three typed relevance questions (instructions and option criteria verbatim from our code) and how each answer is read out for ranking. p_x: probability of option x; $\mathbb{E}[\ell]$: probability-weighted level.

Format	Question and options	Primary score
Choice-3 (C3)	*How relevant is the passage to the search query?* relevant: "The passage directly answers or fully addresses the query."; partially: "The passage is on topic and partly addresses the query, but is incomplete or indirect."; not_relevant: "The passage does not address the query."	$\frac{1}{2}(2p_{\text{rel}} + p_{\text{part}})$
Noul	*Does the passage answer or address the search query?* true: "The passage answers or substantively addresses the query."; false: "The passage is off topic or does not help answer the query."	$P(\text{yes})$
Score-4 (S4)	*How relevant is the passage to the search query?* 0: "Irrelevant: the passage has nothing to do with the query."; 1: "Related: on topic but does not answer the query."; 2: "Highly relevant: partly answers the query."; 3: "Perfectly relevant: dedicated to the query and fully answers it."	$\mathbb{E}[\ell]/3$

over-confident, although self-consistency estimates are better calibrated [40], and fine-tuned small LLMs can be trained to give calibrated relevance for RAG [39]. For deciding how much context a generator sees, learned selectors [37], score-gap heuristics [58], context filtering and compression [9, 66, 68], and retrieval evaluators that trigger corrective actions [70] replace a fixed k; other work decides whether and how often to retrieve at all [2, 28], or interleaves retrieval with reasoning for multi-hop questions [63]. Irrelevant passages and their position affect RAG answers in non-obvious ways [17, 32]. We test whether Platt-scaled decision-model probabilities make such selection a simple threshold.

3 Decision Models as Pointwise Rerankers

For a query q and candidate passage d, we send the state $\{\texttt{query} : q, \texttt{passage} : d\}$ with three typed relevance questions (Table 1). All three can be asked in one request (Jeff's server evaluates each question in a separate forward pass; Laya answers all three in one shared pass), so every decision model in our study answers all three formats in a single call per pair; the three answers are therefore not independent samples.

For every backend, the Score-4 value returned by the model equals the probability-weighted level $\mathbb{E}[\ell]$ computed from its option probabilities (checked on 20,000 answers per model; maximum deviation 0.03, from Jev's two-decimal rounding).

3.1 Probability readout

Calibration needs a probability for the *binarized* relevance a collection uses. The strict readout is p_{rel} (C3), $P(\ell \geq 2)$ (S4), or $P(\text{yes})$ (Noul); the lenient readout, for collections whose lowest relevant grade means *partially* relevant, is $p_{\text{rel}} + p_{\text{part}}$ (C3) or $P(\ell \geq 1)$ (S4). We assign the lenient readout to TREC-COVID, DBPedia, and News, whose label definitions describe the lowest relevant grade as partial, and the strict readout elsewhere. *Disclosure:* this assignment was introduced after we had observed a Choice-3 ECE of 43.9 on TREC-COVID under the strict readout. It is the most favorable readout policy for Jev's Choice-3 answers (mean ECE 14.8, versus 19.2 strict

everywhere and 34.8 lenient everywhere); the Noul readout does not depend on it. We therefore lead with Noul and with the strict-everywhere readout in our calibration claims, treat the per-collection policy as a sensitivity analysis, report every Choice-3 and Score-4 system under the strict readout in Table 3 with the policy as marked sensitivity rows, and give both readouts for every collection in Online Resource 1, Table S2. We separately evaluate source-label transfer grouped by numeric binarization threshold (Experimental Setup, *Metrics and tests*); this grouping does not imply identical relevance semantics or the same adaptation as the readout policy.

Exact score ties are broken by BM25 rank (Online Resource 1, *Setup details*).

4 Experimental Setup

4.1 Collections and candidates

We use the candidates and relevance judgments of the RankGPT/RankLLM evaluation [55, 57] (our rerankers are pointwise, not listwise): TREC DL 2019–2023 passage [12–16] (43/54/53/76/82 queries) and eight BEIR collections [60]—TREC-COVID, NFCorpus, Touche, DBPedia, SciFact, Signal, News, and Robust04 (308 DL and 1,525 BEIR queries, 1,833 in total; 15 NFCorpus queries have no BM25 candidates and score zero for every system). Candidates are the Pyserini [31] BM25 top-100 runs distributed with RankLLM, evaluated with the official Pyserini qrels; our BM25 nDCG@10 reproduces RankGPT's reported values on DL19/20 and seven of the eight BEIR collections (NFCorpus: 32.2 vs. 30.8). Two caveats apply to the newer DL years: 31 of the 82 DL23 queries are synthetic (generated by T5 or GPT-4), and the DL22 judgments were propagated to near-duplicate passages; we follow common practice and keep both. All systems rerank the full top-100 and are evaluated with `ir_measures` [35]. nDCG@10 uses the graded judgments with linear gain; binary measures and calibration count grade ≥ 2 as relevant on TREC DL and grade ≥ 1 on BEIR (judged and unjudged counts per collection are in Online Resource 1, Table S11). We use depth 100 because it is the depth at which RankGPT and RankLLM report results on these runs, and because the strong rerankers still improve between depths 50 and 100 on TREC DL (Online Resource 1, Table S1).

4.2 Systems

Cross-encoders, both trained with MS MARCO [4] data: `ms-marco-MiniLM-L6-v2` (22M) and `bge-reranker-v2-m3` (568M, fp16) [3, 7], both with a 512-token window; we rank by their logit z and use $\sigma(z)$ as their raw probability. Label-probability LLM rerankers, which read relevance from the probability of a "yes" token rather than from a typed answer: Qwen3-Reranker-0.6B, 4B, and 8B [71] with their official instruction template, ranking by the logit $\log P(\text{yes}) - \log P(\text{no})$ and using its sigmoid, $P(\text{yes})$, as probability (4B and 8B with 4-bit NF4 weights [19] to fit an 8 GB GPU), and Qwen3-235B-A22B-Instruct-2507 (Qwen3 family [47]) prompted pointwise for a Yes/No answer, with $P(\text{yes})$ renormalized over the Yes and No tokens among the top-5 next-token probabilities [72] (via OpenRouter; DL19–23, TREC-COVID, NFCorpus, and Robust04 only, for cost). As a same-base control we ask

Clef's base model, Qwen3.8-27B, the pointwise Yes/No prompt (Cloudflare's deployment via OpenRouter, reasoning disabled; on the seven collections where Clef was run), and Jeff's base model, Qwen3.5-0.8B (locally, thinking disabled, on all 13 collections; P(yes) from the full next-token distribution). For Qwen3-235B and Jeff's base model we also use a graded readout [72]: the same prompt asks for a 0–4 rating, and passages are ranked by the expected rating over the digit tokens, renormalized over the digits present (for Qwen3-235B, among the top-5 tokens; at least one digit always appeared and all five in 90% of judgments). All of these read the first 448 passage tokens, the cross-encoders' budget. Qwen3-Reranker-0.6B is close to bge-reranker-v2-m3 on average but significantly worse on DL22, DL23, and Touche and significantly better on TREC-COVID, NFCorpus, and Robust04. Decision models, all used without task-specific tuning on our side: Laya (421M, ModernBERT-large [67]) [11]; Jeff [22] (Jeff-Qwen3.5-0.8B v1.2); Jev 1.13 (snapshot `20260917`) via the OpenRouter Decisions API [64]; and Clef 27B [8], an open-weight decision model adapted from Qwen3.8-27B, through Cloudflare's deployment on the same API (DL19–23, TREC-COVID, and Robust04 only, for cost). Jev is a closed model whose training data we cannot inspect; MS MARCO, BEIR, and Robust04 are old, public collections that may overlap with it. For reference we also quote published results on the identical BM25 top-100 runs: monoT5-3B [42] and RankGPT [57].

4.3 Context budgets

Models read different amounts of each passage: the cross-encoders 512 tokens including the query, Laya about 320 passage tokens, Jev, Clef, and Jeff the first 4,000 characters (≈1,000 tokens), and the label-probability LLMs 448 passage tokens. On TREC DL, DBPedia, and Signal almost no passage is truncated; on the BEIR abstract collections (TREC-COVID, NFCorpus, SciFact) 29–38% exceed the cross-encoders' budget; and on Touche, News, and Robust04 69–97% do (Online Resource 1, Table S12). As newer-content benchmark checks we add NovelEval-2306 (released with RankGPT [57]; 21 queries, 20 passages each, graded 0–2) and FutureQueryEval [1] (148 queries about events after April 2025, graded 0–2, BM25 top-100 built with `bm25s` [33] over its 2,787-document corpus); they are reported separately and not included in the 13-collection averages. For the three long-document collections we add an equal-budget condition in which every passage is cut to its first 448 tokens before any model sees it, and bge-reranker-v2-m3 at a 2,048-token window. On these and the three abstract collections, Qwen3-Reranker-8B also reads the first 4,000 characters of each passage, as the decision models do.

4.4 Metrics and tests

Ranking: nDCG@10 with paired two-sided t-tests over queries; Holm correction [26] covers the collections available for each system–baseline pair, not across the 75 system–baseline pairs. Table averages weight collections equally. We additionally pool collection effects with DerSimonian–Laird random effects [18] and report both normal inference and a post-hoc modified Knapp–Hartung sensitivity analysis [51]. mKH uses Student-t with $k - 1$ degrees of freedom and floors the residual variance multiplier

at one. Pooled TOST [53] with a ±2 nDCG margin is exploratory: the margin was chosen after seeing results, and normal/mKH conclusions are distinguished. We also report Judged@10 and condensed nDCG′@10 [52] to assess sensitivity to incomplete pools. Calibration: ECE (15 equal-width bins) and Brier score [5] on judged pairs from nine collections with negatives (DL19–23, TREC-COVID, DBPedia, News, Robust04), with paired query-cluster bootstrap 95% CIs (2,000 replicates; Holm over nine collections). Transfer Platt scaling uses no target labels: All pools the other collections with equal weight per pair; Group restricts sources to the same numeric binarization threshold and weights source collections equally (DL→DL, dense BEIR→BEIR). Numeric thresholds need not encode identical relevance semantics, especially for partial versus strict relevance in BEIR. In-collection Platt scaling uses target labels with 10× repeated five-fold query cross-fitting and serves as a reference floor. Because judged pairs are not a random sample of candidates, we also report ECE when unjudged candidates are treated as non-relevant (Online Resource 1).

4.5 RQ3: context selection

We use MuSiQue-Ans dev [62] (multi-hop; up to 20 candidate paragraphs per question, 2–4 labelled as supporting) and NQ-open test [30] with the DPR top-20 passages [29] (single-hop; 18.6% of final-set questions have no answer-bearing passage; gold labels come from DPR's string-match `has_answer` flag). With seed 13, each dataset is split into calibration (200), pilot (300), and final (1,000) questions with disjoint IDs. MuSiQue families were not kept together: 671 final questions share a decomposition sub-question ID with a design split. For MuSiQue we therefore built a *family-disjoint* split after the original analysis. Families are connected components of MuSiQue dev questions that share any decomposition ID; we drop every family that contains a pilot question and assign whole families to a calibration split (`calib_fd`, 199 questions) and a final split (`final_fd`, 597 questions in 270 families). Because the large 3- and 4-hop families reach the pilot, `final_fd` is 88% 2-hop (526/58/13 two-, three-, and four-hop questions, versus 52/31/17% in the original final split). We reuse the nine policies and thresholds locked after the original pilot without re-tuning, and refit Platt scaling and Jev's readout on `calib_fd` (construction details in Online Resource 1, *Setup details*). The family-disjoint split is our primary MuSiQue result; the original split and the retrospective 329-question subset without directly shared sub-question IDs are reported as sensitivity analyses. NQ questions have no decompositions and keep the original split. The calibration set is used to fit Platt scaling on gold labels and to choose Jev's probability readout by Brier score (Noul on MuSiQue, Choice-3 strict on NQ); thresholds are therefore applied to *Platt-scaled* probabilities that use target-dataset labels. Jev ranks candidates by its Score-4 expected level, bge-reranker-v2-m3 by its logit. After the final results we added, on the original and family-disjoint MuSiQue splits and on NQ, Qwen3-Reranker-8B (4-bit) and Qwen3-Reranker-0.6B, which rank by their yes/no logit (448-token passage budget, as in RQ1; MuSiQue and DPR passages are shorter); they receive the same nine policies, with Platt scaling fit on the same calibration questions. Policies select passages by fixed top-k, Adaptive-k (cut after the largest gap between consecutive ranking scores [58]—Jev's Score-4 level or the bge or Qwen logit—taking the first of equal gaps, so $1 \leq k \leq 19$), a

threshold $P > t$, or top-2 plus any further passage with $P > t$ ("top-2+threshold"). Ties in a ranking score keep the original candidate order. The nine policies compared between rankers are top-1/2/3/5, Adaptive-k, top-2+threshold at 0.3 and 0.5, and threshold at 0.2 and 0.3. Three generators answer with the selected passages (temperature 0, at most 32 tokens, no option to abstain; prompt in Online Resource 1): Qwen2.5-7B-Instruct [46], Llama-3.1-8B-Instruct [24], and Qwen3-30B-A3B-Instruct-2507, via OpenRouter. We score answers with SQuAD-normalized F1 [49]. Policies and thresholds were fixed after the pilot before final generation. The locked threshold-rule comparisons were against top-3 and Adaptive-k; comparisons against top-2 and top-5 were added later and are exploratory. The pilot motivated the top-2+threshold floor. Raw thresholds (MuSiQue pilot Qwen2.5-7B: 41.6 F1 at 0.3 versus 43.9 Platt-scaled) were dropped from the lock and added after final results as a post-hoc analysis. Their probability readout still uses labelled calibration questions, so they avoid Platt fitting but are not label-free. Comparisons are paired over questions. For each dataset and generator, Holm correction is applied separately to the nine ranker-effect tests (Jev vs. bge-reranker-v2-m3 under each policy) and to the four threshold-rule tests (top-2+threshold at 0.3 vs. top-2, top-3, top-5, and Adaptive-k); an earlier version of our analysis mixed the two families, and we separated them after an internal review. We add 95% CIs and, after seeing the final results, TOST equivalence tests with a ± 2 F1 margin, which we report as post-hoc and without multiplicity correction. Question-level tests do not account for dependence between questions of the same MuSiQue family. Our primary MuSiQue tests are therefore family cluster-robust paired t-tests (CR1, $G-1$ degrees of freedom for G families), Holm-corrected over the nine policies as above; question-level tests, a family sign-flip randomization test, and family-bootstrap 95% CIs are sensitivity analyses (Online Resource 1, *Setup details*). NQ questions have no families, so NQ keeps question-level tests.

4.6 Hardware

Local models ran on a laptop RTX 4060 (8 GB); Jeff ran through its reference server, whose attention kernels fall back to a slow reference implementation on Windows, so its latency is pessimistic. API models were called through OpenRouter; Jev received all three questions per call. All API calls in this study cost about $55 in total, so every experiment can be reproduced on a small budget. OpenRouter routed the Llama and Qwen3-30B generators across several hosting providers (Qwen2.5-7B was served by one), which may serve differently quantized weights; provider assignment was roughly similar across policies within a run, but the Qwen3-Reranker contexts were generated later with a different provider mix, so comparisons with them also differ in serving provider; we record the provider of every call.

4.7 Use of generative AI

The authors used Claude (Anthropic), through the Claude Code tool, to help write and debug analysis code, run and audit experiments, and edit the manuscript text. The authors reviewed and verified all outputs and take full responsibility for the content

of this article. The language models evaluated in the study are described above and are separate from this assistance.

5 Results

5.1 RQ1: Effectiveness

Large decision models are on par with the strongest label-probability reranker run on all 13 collections. Over the 13 collections (Table 2; all formats in Online Resource 1, Table S8), Jev averages 55.3–56.1 nDCG@10 across its three formats (Score-4: 56.1), Qwen3-Reranker-8B 56.0, Qwen3-Reranker-4B 55.7, bge-reranker-v2-m3 53.7, Qwen3-Reranker-0.6B 52.9, MiniLM 51.9, and BM25 41.9. Pooling per-collection effects with random effects, Jev S4 minus Qwen8B is +0.3 (90% CI $[-1.3, 1.9]$), and Jev S4 minus bge is +2.6 (95% CI $[0.1, 5.1]$). Collection heterogeneity is large ($I^2 \approx 0.9$); with the small-sample modified Knapp–Hartung (mKH) correction both intervals widen slightly ($[-1.4, 2.0]$ and $[-0.4, 5.6]$; Online Resource 1, Table S7), so we read the evidence as parity with Qwen8B (equivalence within ±2 points holds under normal inference, TOST $p = 0.04$, and the mKH interval reaches the margin) and a likely but not conclusive pooled advantage over bge. Because effects vary across collections, a new collection could favor either system: 95% prediction intervals are $[-6.9, 7.5]$ for Jev S4 minus Qwen8B and $[-7.4, 12.6]$ for Jev S4 minus bge. All Qwen8B results use the 4-bit model reading 448 passage tokens; the Long documents subsection tests a 4,000-character budget. Without Robust04, Jev S4 minus Qwen8B is −0.3 (90% CI $[-1.4, 0.8]$; mKH $[-1.6, 1.0]$), so the parity does not depend on that collection. At the collection level, Jev S4 is significantly better than bge on five collections (DL21, TREC-COVID, NFCorpus, SciFact, Robust04) and significantly worse on none; against Qwen8B it is better on Robust04 (+7.9) and worse on Signal (−3.9, Holm). No single system wins everywhere: Clef is best on DL20, DL23, and Robust04, Jev on DL21, NFCorpus, and SciFact, the Qwen rerankers on DL19, TREC-COVID, DBPedia, and News, bge on DL22, MiniLM on Signal, and BM25 on Touche, where every neural reranker falls below the first stage. On TREC DL alone, Jev S4 averages 61.1 versus bge 60.6 and MiniLM 59.6 (pooled Jev–bge +0.7, 90% CI $[-1.2, 2.6]$); on BEIR, 53.0 versus 49.3. These results use BM25 candidates. On SPLADE++ [23] candidates for DL19, Jev scores 3.1–5.5 points below bge, but more of its top 10 are unjudged (Judged@10 87% versus 95%); with unjudged candidates removed, the gap is 0.1–1.7 points and not significant. On DL20, Jev Noul is the best system under either scoring (Online Resource 1, *Neural first stage*). Neither scoring bounds the difference under complete judgments: over all possible grades of the unjudged passages in the two top-10 lists, Jev S4 minus bge on DL19 could lie anywhere between −8.7 and +7.8. The DL19 deficit therefore cannot be separated from incomplete judgments, and whether the DL similarity holds on a strong neural first stage remains open.

Robust04 contributes much of the margin. Without it, Jev and bge average 55.3 and 53.8 over 12 collections (pooled +1.7, 95% CI $[0.1, 3.4]$; mKH $[-0.4, 3.9]$). Jev, Clef, Qwen235B, and the three Qwen3-Rerankers are unusually strong on Robust04 (Jev 65.9, Clef 66.5, Qwen235B 60.2, Qwen0.6B/4B/8B 55.8/57.4/58.0, versus bge 51.9), which is consistent with either an advantage on long news articles or familiarity

Table 2 nDCG@10 for reranking the BM25 top-100 (Pyserini runs and qrels, as in RankGPT/RankLLM). $^{\uparrow}/^{\downarrow}$: significantly better/worse than bge-reranker-v2-m3 (paired t-test over queries, Holm-corrected over the collections each pair was evaluated on, $p < 0.05$). Best displayed configuration per column in bold; all formats and controls appear in Online Resource 1, Table S8. Averages are reported only for systems run on every collection of the group (dash otherwise); matched comparisons for Clef and the P(yes) LLMs are given in the text. All: average over all 13 collections. Touche, News and Robust04 contain long documents that every model truncates (Online Resource 1, Table S12).

	(a) TREC Deep Learning						
System	DL19	DL20	DL21	DL22	DL23	*Avg*	**All**
BM25 (first stage)	50.6	48.0	44.6	26.9	26.2	39.3	41.9
Cross-encoders							
MiniLM-L6 (22M)	72.7	67.5	65.6	49.9	42.2	59.6	51.9
bge-reranker-v2-m3 (568M)	72.4	68.1	66.2	**51.4**	45.1	60.6	53.7
Label-probability LLM rerankers							
Qwen3-Reranker-0.6B	69.4	64.0	65.1	46.3$^{\downarrow}$	38.8$^{\downarrow}$	56.7	52.9
Qwen3-Reranker-4B (4-bit)	**74.0**	68.2	66.2	50.4	45.0	60.7	55.7
Qwen3-Reranker-8B (4-bit)	73.6	66.5	66.2	51.1	44.9	60.5	56.0
Qwen3-235B, P(yes)	68.6	66.5	65.6	46.2	42.8	57.9	–
Qwen3.8-27B (Clef's base), P(yes)	73.0	69.8	69.2	48.8	45.5	61.2	–
Qwen3.5-0.8B (Jeff's base), P(yes)	42.9$^{\downarrow}$	38.9$^{\downarrow}$	50.6$^{\downarrow}$	26.3$^{\downarrow}$	28.6$^{\downarrow}$	37.4	31.8
Laya (421M)							
Noul	57.1$^{\downarrow}$	52.1$^{\downarrow}$	55.0$^{\downarrow}$	36.2$^{\downarrow}$	32.0$^{\downarrow}$	46.5	40.3
Jeff (0.8B)							
Noul	59.5$^{\downarrow}$	50.1$^{\downarrow}$	52.5$^{\downarrow}$	33.0$^{\downarrow}$	30.4$^{\downarrow}$	45.1	39.4
Clef 27B (open weights, API; 7 collections)							
Score-4	72.2	**70.2**	69.3	49.1	**46.8**	**61.5**	–
Jev 1.13 (API)							
Noul	71.3	70.0	69.5	50.1	45.0	61.2	56.0
Score-4	72.8	69.9	**70.0**$^{\uparrow}$	48.5	44.3	61.1	**56.1**

	(b) BEIR								
System	Covid	NFC	Touche	DBP	SciF	Signal	News	Rob04	*Avg*
BM25 (first stage)	59.5	32.2	**44.2**	31.8	67.9	33.1	39.5	40.7	43.6
Cross-encoders									
MiniLM-L6 (22M)	74.5	34.9	28.4$^{\downarrow}$	43.9	69.5$^{\downarrow}$	**33.9**$^{\uparrow}$	44.3	47.3$^{\downarrow}$	47.1
bge-reranker-v2-m3 (568M)	78.3	34.3	38.3	44.0	75.0	29.0	43.6	51.9	49.3
Label-probability LLM rerankers									
Qwen3-Reranker-0.6B	**86.2**$^{\uparrow}$	36.8$^{\uparrow}$	29.2$^{\downarrow}$	43.3	76.8	30.6	45.5	55.8$^{\uparrow}$	50.5
Qwen3-Reranker-4B (4-bit)	84.1$^{\uparrow}$	37.1$^{\uparrow}$	34.9	**46.4**$^{\uparrow}$	77.8	33.8$^{\uparrow}$	48.6$^{\uparrow}$	57.4$^{\uparrow}$	52.5
Qwen3-Reranker-8B (4-bit)	83.8$^{\uparrow}$	37.5$^{\uparrow}$	36.2	45.5	79.2$^{\uparrow}$	33.7$^{\uparrow}$	**51.6**$^{\uparrow}$	58.0$^{\uparrow}$	**53.2**
Qwen3-235B, P(yes)	80.0	34.6	–	–	–	–	–	60.2$^{\uparrow}$	–
Qwen3.8-27B (Clef's base), P(yes)	83.7	–	–	–	–	–	–	62.5$^{\uparrow}$	–
Qwen3.5-0.8B (Jeff's base), P(yes)	60.2$^{\downarrow}$	17.3$^{\downarrow}$	19.1$^{\downarrow}$	21.4$^{\downarrow}$	28.0$^{\downarrow}$	24.4$^{\downarrow}$	22.7$^{\downarrow}$	33.5$^{\downarrow}$	28.3
Laya (421M)									
Noul	74.2	28.9$^{\downarrow}$	22.5$^{\downarrow}$	28.7$^{\downarrow}$	44.5$^{\downarrow}$	21.8$^{\downarrow}$	35.7$^{\downarrow}$	34.5$^{\downarrow}$	36.4
Jeff (0.8B)									
Noul	67.3$^{\downarrow}$	24.5$^{\downarrow}$	23.5$^{\downarrow}$	30.0$^{\downarrow}$	46.7$^{\downarrow}$	27.4	33.6$^{\downarrow}$	34.1$^{\downarrow}$	35.9
Clef 27B (open weights, API; 7 collections)									
Score-4	84.0	–	–	–	–	–	–	**66.5**$^{\uparrow}$	–
Jev 1.13 (API)									
Noul	84.1	37.7$^{\uparrow}$	35.5	42.7	79.8$^{\uparrow}$	28.3	47.6	66.0$^{\uparrow}$	52.7
Score-4	85.0$^{\uparrow}$	**38.4**$^{\uparrow}$	31.6	44.5	**80.0**$^{\uparrow}$	29.8	48.6	65.9$^{\uparrow}$	53.0

with these public 2004 topics. On newer-content benchmarks, Jev scores 62.5–63.5 on FutureQueryEval versus bge 58.2 ($p \leq 0.001$) and 90.4–91.3 on NovelEval versus 81.8 ($p \approx 0.05$); its leads over Qwen8B (61.2 and 86.2) are not significant.

Published rerankers. On identical candidates, published monoT5, DuoT5, and RankZephyr scores are within 1.6 nDCG@10 points of Jev S4 on TREC DL and BEIR; because published per-query outputs are unavailable, these comparisons are descriptive (Online Resource 1, *Comparison with published rerankers*).

Clef and the larger yes/no baseline. On the seven collections where Clef was run, Clef averages 64.4–65.4 across formats, Jev 64.1–65.2, and bge 61.9; Jev S4 minus Clef S4 is within ±2 (normal pooled 95% CI $[-1.0, 0.4]$; also under mKH). Qwen235B's $P(\text{yes})$ averages 58.1 on its eight collections versus Jev Noul 61.7 (Jev significantly better on five, worse on none); a graded 0–4 readout raises Qwen235B to 59.8 (+1.8, 95% CI $[0.8, 2.8]$), still below Jev S4 at 61.8.

Same-base controls: Clef is close to its base LLM; Jeff improves on its base. On the seven collections where Clef was run, Clef's base Qwen3.8-27B averages 64.6 as a $P(\text{yes})$ ranker versus Clef's 64.4–65.4, and itself exceeds Qwen235B by 2.3–4.4 points on each. On DL19–23 and TREC-COVID, none of the 18 collection-level differences is significant (all within 2.5 points); on Robust04, every Clef format is 3.7–4.0 points ahead (Holm $p < 10^{-5}$), where Clef reads 4,000 characters but its base only 448 tokens (92% of Robust04 candidates are longer; Online Resource 1, Table S12). Pooled over the seven collections, Clef S4 minus its base is +1.0 (90% CI $[-0.3, 2.2]$; mKH $[-0.5, 2.5]$; Online Resource 1, Table S7), so ±2 equivalence is not established. Outside Robust04 we therefore find no ranking benefit of Clef over its base, in line with the early audit's caution about attributing quality to typed readouts [59]. In contrast, Jeff's formats average 39.1–39.4 over 13 collections versus 31.8 for its base (graded: 26.4), significantly better on 7–8 collections and worse on none. Because prompts and context budgets differ between each model and its base, these are configuration comparisons rather than isolated training effects. Jev's base, size, and training data are unknown; its 4,000-character context also exceeds the 448-token budget of the label-probability controls, and on Robust04 its Noul lead over Qwen235B shrinks from 5.7 to 2.4 at equal passage budget.

Small decision models are weak deep rerankers. Laya and Jeff average 39.1–41.2, below BM25's 41.9 at depth 100. They retrieve more unjudged passages (DL Judged@10 76–81% versus bge 96%) and score higher on condensed nDCG′ (58.6/57.3 versus BM25 51.8 and bge 64.0), and Laya's BEIR nDCG falls from 43.7 at depth 10 to 37.4 at depth 100 while its condensed nDCG rises (60.4 to 61.9). Part of their deficit may therefore reflect incomplete pools, although removing unjudged passages cannot show that those passages are relevant. Jev retrieves few unjudged passages (DL Judged@10 90.0%; condensed nDCG 66.9 versus bge 64.0). Within each family, the three question formats differ by at most 1.3 average points, much less than the differences between families.

5.2 Long documents

At the same first 448 passage tokens, Jev S4 exceeds bge by +11.5 on Robust04 and +4.9 on News ($p < 0.01$, Holm over three collections), versus +13.9 and +5.0

with full context, and loses on Touche (−9.9; Online Resource 1, Table S12). The context budget does not explain the Jev–Qwen8B comparison either: whether Jev reads Qwen8B's 448 tokens or Qwen8B reads Jev's 4,000 characters, Jev's Robust04 lead holds and Qwen8B's Touche lead widens, and on the three abstract collections Qwen8B's nDCG@10 changes by at most 0.1 point with 4,000 characters. Substituting the 4,000-character runs on these six collections moves the pooled Jev S4 minus Qwen8B over all 13 from +0.3 to −0.1 (95% CI $[-1.7, 1.6]$; per-collection results in Online Resource 1, *Long documents*). Giving bge-reranker-v2-m3 a 2,048-token window, which reads most documents in full, essentially closes the gap on News (47.8) but does not help on Robust04 (51.6 vs. 51.9).

5.3 RQ2: Calibration

Raw cross-encoder scores were never trained to be probabilities, so a raw comparison mainly asks whether a model can be thresholded *out of the box*. The fairer comparison is with *transfer* calibration, which fits Platt scaling on other collections and uses no labels of the target collection (Table 3). We lead with the Noul readout, which needs no strict/lenient choice, and with the strict readout applied everywhere; the per-collection readout policy, chosen after observing TREC-COVID (*Probability readout*), is reported as a sensitivity analysis.

Out of the box, Jev and Laya yes/no probabilities are better calibrated than raw cross-encoder scores. Jev Noul has mean ECE 5.0 points below bge (paired query-bootstrap 95% CI $[-6.3, -3.9]$), lower on eight of nine collections, significantly lower on five after Holm, and significantly higher on none (TREC-COVID is slightly higher, +0.17). Laya Noul is 9.2 points lower ($[-10.8, -7.5]$). Jeff Noul is the exception (25.6 versus bge 20.0). On DL, the trained rerankers over-predict relevance (ECE 51.1/36.5/27.6 for Qwen0.6B/4B/8B, 47.0 for MiniLM, and 23.1 for bge; Online Resource 1, Fig. S1). The advantage depends on the readout for the graded formats. With the strict readout everywhere, Jev C3 is only 0.7 points below bge ($[-2.5, 0.9]$) and Jev S4 2.1 points below ($[-3.4, -0.9]$); under the per-collection policy, Jev C3 is 5.1 points below ($[-7.1, -3.4]$), while Jev S4 over-predicts relevance (mean prediction 0.55 against a base rate of 0.31) because the lenient readout $P(\ell \geq 1)$ counts "related but not answering" as relevant. Jev C3's DBPedia ECE is worse than bge's by 9.9 under the policy and 14.1 under strict, and Robust04 by 0.9 under both. These differences change by at most 0.4 points with 5–30 bins or equal-mass binning. A Brier decomposition attributes Jev Noul's advantage to both reliability (0.043 vs. 0.067) and resolution (0.066 vs. 0.045).

Same-threshold transfer calibration closes the gap. Pooling all other collections (pair-weighted) yields ECE 14.9 for bge and 16.8 for MiniLM, versus Jev C3/S4/Noul 15.1/14.8/14.2 and Laya C3 11.0 under the strict readout (9.1/8.2 for Jev C3/S4 and 7.5 for Laya C3 under the per-collection policy); collection-balanced pooling changes little. Restricting sources to collections with the same numeric relevance threshold (DL→DL and BEIR→BEIR, collections weighted equally) yields bge 6.9, essentially equal to Jev Noul at 6.8, slightly below Jev S4 (7.5), and below Jev C3 (9.1) and Laya C3 (9.6). The BEIR group mixes partial-relevance definitions with strict Robust04, so it approximates rather than matches target relevance semantics. The practical

Table 3 Calibration of P(relevant) on the nine collections with judged negatives (DL19–23, TREC-COVID, DBPedia, News, Robust04; unweighted mean over collections, ECE×100, 15 bins, judged pairs only). Raw: no recalibration. Transfer (no target labels): Platt scaling fit on the other eight collections pooled (All) or only on the same numeric-threshold group, balancing source collections (Group: DL→DL, BEIR→BEIR); the group need not share identical relevance semantics. All weights pairs equally. In-coll.: Platt with 10× repeated 5-fold query cross-fitting on the target collection (needs target labels; SD over repetitions ≤0.6 on average, at most 1.7 on one collection). C3/S4: Choice-3/Score-4, strict readout on every collection. †Sensitivity row: post-hoc per-collection readout policy (lenient on TREC-COVID, DBPedia, News; see *Probability readout*); excluded from bold. *Seven of the nine collections (no DBPedia, News). Bold: best among systems on all nine.

	ECE					
		Transfer				
System	Raw	All	Group	In-coll.	Brier	$\bar{p}$
MiniLM-L6	35.4	16.8	8.7	3.0	0.345	0.58
bge-v2-m3	20.0	14.9	6.9	2.6	0.215	0.39
Qwen3-Rer.-0.6B	36.1	17.9	9.6	2.4	0.357	0.62
Qwen3-Rer.-4B	26.9	19.0	9.2	2.4	0.265	0.47
Qwen3-Rer.-8B	21.6	17.5	9.6	2.5	0.221	0.41
Qwen3-235B P(yes)*	20.5	10.5	9.5	3.1	0.211	0.39
Laya C3	12.2	11.0	9.6	2.5	0.189	0.25
policy†	11.0	7.5	9.3	2.3	0.184	0.36
Laya Noul	**10.7**	**10.7**	9.5	2.2	0.185	0.35
Laya S4	17.9	11.1	9.3	2.2	0.212	0.47
Jeff C3	13.9	13.5	10.8	2.2	0.194	0.35
policy†	15.3	5.2	5.9	2.5	0.195	0.46
Jeff Noul	25.6	11.7	8.6	**2.2**	0.253	0.56
Jeff S4	15.3	14.1	9.7	2.3	0.196	0.40
Clef C3*	12.3	14.4	10.8	4.2	0.167	0.21
policy*†	7.8	8.3	10.4	3.3	0.134	0.28
Clef Noul*	12.5	12.4	7.3	2.7	0.146	0.36
Clef S4*	11.8	13.0	7.1	2.8	0.138	0.35
policy*†	11.6	8.9	9.1	2.7	0.139	0.40
Jev C3	19.2	15.1	9.1	3.6	0.206	0.18
policy†	14.8	9.1	8.9	4.4	0.169	0.36
Jev Noul	15.0	14.2	**6.8**	3.0	**0.170**	0.38
Jev S4	17.9	14.8	7.5	5.5	0.192	0.41
policy†	25.4	8.2	7.9	5.0	0.240	0.55
Base rate of relevance (mean over the nine collections): 0.31						

advantage of decision-model probabilities is therefore that they need no labelled source collections, not that they are better calibrated than a cross-encoder with a cheap same-threshold transfer fit.

Raw calibration depends on the prompt wording. Rewording the question while keeping its meaning moves raw ECE far more than ranking (Online Resource 1, *Prompt sensitivity*): for Jev Noul, three paraphrases give 7.5–25.9 on DL19 (original

12.4) and 14.3–34.5 on DL20 (20.7), while nDCG@10 stays within 2.7 points; Qwen3-Reranker-0.6B's default instruction is its least calibrated wording, and Qwen3-235B's original prompt its best calibrated. The 5-point raw-ECE gap between Jev Noul and bge is thus smaller than the spread that rewording alone produces, and its direction could change under another prompt. We report it as a property of the prompts we used, not of decision models in general.

Other models. The same-base controls are calibrated much like their bases (Noul ECE 12.5 for Clef versus 12.1 for its base, 25.6 for Jeff versus 26.5); Clef's graded formats and Qwen3-235B are reported in Online Resource 1, *Calibration of other models*.

Pool dependence. Calibration is measured on judged candidates, whose mean base rate is 0.31. Treating unjudged candidates as non-relevant preserves the raw advantage of Jev Noul and Laya Noul but reverses that of Jeff C3 under either readout (Online Resource 1, Table S3).

5.4 Efficiency

Per query–passage pair, the local cross-encoders and Qwen3-Reranker-0.6B take 0.3–14 ms (median) on a laptop GPU, Qwen3-Reranker-4B and 8B 102 and 148 ms (4-bit), Laya 58 ms, and Jeff 100 ms (with slow reference kernels); the API models take a median 176 ms (Jev), 401 ms (Clef), and 850 ms (Qwen3-235B) per call with 8–24 concurrent requests. Reranking 100 passages costs $3.4 per 1k queries with Jev on average ($2.4–5.7; long documents cost more) and $2.3 with the Qwen3-235B yes/no baseline; on the seven collections where both were run, Clef 27B costs about 4.8 times as much as Jev ($15.3 vs. $3.2). Local and API latencies come from different hardware, serving stacks, and concurrency levels, so they indicate orders of magnitude rather than a controlled speed comparison.

5.5 RQ3: Context selection

Table 4 reports answer F1 on the 597 family-disjoint MuSiQue final questions, our primary multi-hop result; Online Resource 1, Table S13 reports the original 1,000-question final splits of MuSiQue and NQ.

Better ranking improves multi-hop answers. On the family-disjoint MuSiQue split, passages selected by Jev beat those selected by bge-reranker-v2-m3 under every policy and with every generator, by +2.8 to +16.9 F1. Twenty-three of the 27 comparisons (nine policies, three generators) pass the family-clustered Holm test (Online Resource 1, Table S9; 26 pass question-level Holm, and 26 have family-bootstrap 95% intervals excluding zero). The four that do not are top-3 and top-5 with Qwen2.5-7B (Holm $p = 0.06$) and top-5 and $P > 0.2$ with Qwen3-30B; only top-5 with Qwen3-30B ($+2.8$, $[-0.6, 6.1]$) has a family-bootstrap interval that includes zero. Exact match agrees in direction: Jev's EM exceeds bge's in all 27 comparisons, 18 of them significant with family clustering. The gains are largest when few passages are selected: top-1 +12.1 to +16.9 and Adaptive-k +11.8 to +12.7, against +5.2 to +7.8 for top-3 and +2.8 to +3.6 for top-5. With every generator, Jev's top-3 (45.5, 42.3, 51.8) is at least as good as giving the generator all 20 candidates (40.8, 38.5, 51.1) with a fifth of the

Table 4 RQ3 on the family-disjoint MuSiQue split: answer F1 on the 597 final questions (88% 2-hop) whose decomposition families share no sub-question with the calibration split or the original pilot. Policies and thresholds are those locked after the original pilot; Platt scaling and Jev's readout are refit on the family-disjoint calibration split. †: Jev significantly better than bge-reranker-v2-m3 under the same policy (family cluster-robust paired t-test, Holm-corrected over the nine policies, $p < 0.05$; Online Resource 1, Table S9). ‡: Jev significantly better than Qwen3-Reranker-8B under the same policy (same test and correction, added after final results).

Policy	MuSiQue Qwen2.5-7B	MuSiQue Llama-3.1	MuSiQue Qwen3-30B	Avg. n
No context	13.5	10.4	20.2	0.0
All 20 passages	40.8	38.5	51.1	20.0
Gold passages (oracle)	62.0	58.7	68.6	2.1
Jev 1.13				
top-1	39.7†‡	33.0†‡	42.1†‡	1.0
top-3	45.5‡	42.3†	51.8†	3.0
top-5	44.2	40.4†	51.7	5.0
adaptive-k	42.6†‡	39.1†‡	48.9†‡	2.3
$P > 0.3$	38.2†	35.6†	44.0†	1.7
top-2 + $P > 0.3$	44.0†‡	40.5†	49.8†	2.5
bge-reranker-v2-m3				
top-1	22.8	20.9	28.8	1.0
top-3	40.3	34.4	46.0	3.0
adaptive-k	30.0	27.3	36.2	3.0
top-2 + $P > 0.3$	36.3	31.8	43.2	2.8
Qwen3-Reranker-8B (4-bit)				
top-1	30.0	26.7	35.7	1.0
top-3	39.8	38.0	46.8	3.0
adaptive-k	35.2	34.0	41.9	3.4
top-2 + $P > 0.3$	39.4	37.0	47.2	3.0
Qwen3-Reranker-0.6B				
top-1	23.8	20.9	27.8	1.0
top-3	35.0	32.7	41.6	3.0
adaptive-k	31.6	27.8	36.8	3.2
top-2 + $P > 0.3$	33.5	29.8	37.6	2.7

generator input tokens (491 vs. 2,676 for Qwen2.5-7B, excluding the cost of scoring the candidates). On NQ, despite Jev's higher recall of answer-bearing passages (R@3 58.4 vs. 52.5; Online Resource 1, Table S4), only one of 27 comparisons is significant.

Sensitivity: the original split and hop count. On the original MuSiQue split, whose final questions share decomposition sub-questions with the design splits, Jev's gains over bge are somewhat larger (+2.6 to +23.2 F1) and all 27 comparisons pass question-level Holm. The gains are larger on 3- and 4-hop than on 2-hop questions, on the original split and among the 58 three-hop questions of the family-disjoint split (Online Resource 1, *RQ3 on the original split and by hop count*).

The gain is not only that of a stronger reranker. On the original split (Online Resource 1, Table S13), Qwen3-Reranker-8B, which matches Jev in RQ1, recovers part of Jev's MuSiQue advantage over bge, but Jev beats it under every policy with every generator (+1.6 to +14.5 F1; 19 of 27 comparisons pass the family-clustered test), and on NQ only at top-1 (+3.1 to +4.8); these comparisons were added after the final results (Online Resource 1, *RQ3 on the original split and by hop count*). On the family-disjoint split (Table 4) the margin over Qwen8B is smaller but keeps its direction. Qwen8B again recovers part of the gap to bge (top-1: 30.0, 26.7, and 35.7 F1 versus bge's 22.8, 20.9, and 28.8), and Jev is ahead in 26 of 27 comparisons (−0.2 to +9.7 F1; top-1 +6.4 to +9.7, Adaptive-k +5.1 to +7.4). Twelve pass the family-clustered

Holm test (14 question-level), 16 have family-bootstrap intervals excluding zero, and none favors Qwen8B; most of the others are top-5 or pure-threshold policies. In exact match Jev is ahead of Qwen8B in 25 of 27 comparisons, five of them significant. Qwen3-Reranker-0.6B is below Jev in all 27 (+5.1 to +15.9).

Jev ranks the answer-bearing hop first. On the original split, the ranking advantage coincides with a difference in *which* passage each ranker puts first. bge-reranker-v2-m3's top passage is a supporting paragraph slightly more often (88.0% vs. 82.2%, McNemar $p < 10^{-4}$), but usually the first hop—the one that shares the question's entities (61.9%)—whereas Jev's top passage is the last, answer-bearing hop in 51.2% of questions against 15.2% ($p < 10^{-80}$; Online Resource 1, Table S4), and 51–60% across Jev's three readouts. The pattern replicates on the family-disjoint split: Jev S4's top passage is the last hop in 46.1% of questions versus 15.9% for bge and 29.0% for Qwen8B (family-clustered differences +30.2, 95% CI [24.3, 36.0], and +17.1, [11.6, 22.6]) and contains the answer in 50.6% versus 22.1% and 34.8%, while bge and Qwen8B put some supporting paragraph first slightly more often (87.4% and 88.3% vs. 82.9%). This preference is not reproduced by rewording a label-probability reranker's instruction: asking Qwen0.6B whether the passage is relevant, answers, or contains the answer yields only 16.5–18.4% last-hop@1 and 51–56% first-hop@1. Qwen3-Reranker-8B lies in between: its top passage is the last hop in 30.2% of questions and the first hop in 44.5%, and contains the answer in 36.6% (Jev 56.5%, bge 22.3%). Its top-3 recall of supporting paragraphs equals Jev's (68.6% vs. 68.7%), yet Jev's top-3 contexts score 2.8–5.3 F1 higher, consistent with which supporting paragraph is selected mattering more than how many. Replacing gold-answer strings with `[MASKED]` reduces Jev S4 last-hop@1 to 43.8% (Noul: 50.3%) from 51.2% (56.4%); masked bge scores 13.4%. Because the marker itself reveals answer location (it appears in 999 of 1,000 last-hop paragraphs but only 1,621 of 19,988 candidates), this perturbation cannot separate question resolution from that cue, prior entity knowledge, or memorized chains.

Thresholds on probabilities do not beat fixed k. A pure threshold at 0.3 on Platt-scaled probabilities leaves the generator with no passage for 10.3% of MuSiQue and 21.6% of NQ questions, which then fall back to closed-book answering (the prompt does not allow abstention). Raw $P > 0.3$ is within 1.3 F1 of the Platt threshold and of top-3 on NQ for every generator; on MuSiQue it selects nothing for 25% of questions and is 3.4–5.9 F1 below top-3 (Online Resource 1, *Raw-probability thresholds*). The top-2+threshold rule is within ±2 F1 (TOST) of top-2, top-3, top-5, and Adaptive-k on NQ for every generator. On MuSiQue it is equivalent to top-2 (and, for Llama, to top-3), but pooled over generators it is 1.9 F1 below Adaptive-k (95% CI $[-3.4, -0.3]$; Holm-adjusted $p = 0.065$), and individual comparisons against Adaptive-k, top-3, and top-5 trend lower by 1.9–2.7 F1 (unadjusted $p = 0.01$–0.05, none significant after Holm). On the family-disjoint split it is 1.5–2.0 F1 below top-3 and 0.9–1.4 F1 above Adaptive-k, with no comparison passing Holm and every family interval crossing zero; it is equivalent to top-2 for Llama and Qwen3-30B and pooled over generators (TOST $p < 0.05$). A pure Platt threshold at 0.3 leaves 16.2% of these questions without context. Because it selects more than two passages for 39% of MuSiQue and 45% of NQ questions (3.0 and 3.5 on average), it also saves no tokens over top-3. Better calibration does not rescue thresholding. On the RAG final sets

Jev's probabilities are not the better calibrated ones (Online Resource 1, Table S10): after Platt scaling fit on the calibration split, ECE on the family-disjoint final pairs is 3.1 for Jev Noul (question-bootstrap 95% CI $[2.8, 3.7]$) versus 1.3 ($[0.9, 1.9]$) for bge and 1.4 for Qwen8B, and on NQ 3.8 for Jev Choice-3 versus 1.4 for bge (Jev Noul, not selected by the Brier criterion, reaches 1.2). Jev also ranks by Score-4 but thresholds another readout, so its passages with $P > 0.3$ are not a prefix of its ranking for 10.7% of family-disjoint MuSiQue questions (21.8% on the original split, 5.0% on NQ). Yet the better-calibrated cross-encoder probabilities fare no better: compared with fixed top-k interpolated to the same average number of passages, $P > 0.3$ on the family-disjoint split is 2.4–6.5 F1 lower for bge and 1.8–4.7 for Qwen8B with every generator, against 2.4–4.0 for Jev, whose top-2+threshold rule is within 0.8 F1; on NQ the thresholds of Jev, bge, and Qwen8B are within 1.5 F1 of it. Jev's practical advantage for RAG is therefore its ranking, not thresholding on its probabilities.

6 Threats to Validity

6.1 Pre-specified versus post-hoc analyses

Several analyses were added or changed after results were observed, and we flag them here in one place. (i) The per-collection strict/lenient readout policy was introduced after observing a Choice-3 ECE of 43.9 on TREC-COVID; we therefore lead with the policy-free Noul readout and the strict-everywhere readout, and give both readouts for every collection (Online Resource 1, Table S2). (ii) The ± 2 nDCG and ± 2 F1 equivalence margins, the TOST analyses, and the modified Knapp–Hartung sensitivity analysis are post-hoc and exploratory. (iii) In RQ3, threshold-rule comparisons against top-3 and Adaptive-k were fixed after the pilot; comparisons against top-2 and top-5 and the raw (non-Platt) thresholds were added after final results. (iv) The 329-question MuSiQue subset is a retrospective sensitivity analysis on existing outputs, not a new confirmatory split. (v) Added during revision, after the original results were known: the family-disjoint MuSiQue split, the downstream comparison with Qwen3-Reranker-8B/0.6B, the strict readout as the primary calibration row for every Choice-3 and Score-4 system, the extension of the Clef base control from three to seven collections, the 4,000-character Qwen3-Reranker-8B control, the prompt paraphrases, the SPLADE++ first stage and its pool-coverage check, the per-hop F1 breakdown, and the exact-match tables. The policies, thresholds, and paraphrases were fixed before the corresponding runs, but the decision to run each analysis was not. (vi) Added in response to peer review, from existing outputs or local models: family-clustered tests as the primary MuSiQue inference, the completion range of the DL19 SPLADE++ difference, prediction intervals, the held-out calibration of the RAG probabilities, the rank/threshold mismatch and equal-size top-k comparison, the judged-count table, the family-disjoint mechanism, exact-match, and hop analyses, the threshold comparison for every ranker, and the 4,000-character Qwen3-Reranker-8B runs on TREC-COVID, NFCorpus, and SciFact.

6.2 Statistical dependence

The three question formats of each decision model are answered in one call per pair and are not independent samples. Holm correction is applied within each system–baseline pair across collections, not across the 22 systems and formats (75 system–baseline pairs). Our primary family-disjoint split removes the sub-question overlap of the original MuSiQue split, and our primary MuSiQue tests cluster questions by family, but the split was built after the original results were seen, and excluding every family that reaches the pilot leaves a split that is 88% 2-hop, so its results speak mainly to 2-hop questions.

6.3 Configurations rather than isolated effects

Same-base comparisons differ in prompts and context budgets (4,000 characters for Jev, Clef, and Jeff; 448 tokens for the label-probability controls), so they compare configurations rather than isolate decision-model training. Qwen3-Reranker-4B/8B use 4-bit weights. Clef and Qwen3-235B cover subsets of collections, and the Clef base control covers Clef's seven. The main results use a single prompt per format, and raw calibration error depends strongly on its wording (Online Resource 1, *Prompt sensitivity*). Latencies come from different hardware and serving stacks.

6.4 Benchmarks and contamination

Our main experiments use BM25 candidates; on SPLADE++ candidates Jev's DL19 deficit to the cross-encoders cannot be separated from incomplete judgments, so first-stage effects are confounded with pool coverage (Online Resource 1, *Neural first stage*). Calibration is measured on judged candidates, and numeric-threshold transfer groups need not share identical relevance semantics. We retain the DL23 synthetic queries and DL22 propagated judgments. Jev is a closed model whose size, training data, and cutoff are unknown; Robust04, MuSiQue, NovelEval, and FutureQueryEval all predate its release, so contamination cannot be excluded, and the answer-masking perturbation cannot separate reasoning from memorization. The RAG comparison covers bge-reranker-v2-m3 and Qwen3-Reranker-8B/0.6B but not the API label-probability baselines, the Qwen comparisons were added after the final results, thresholds use target calibration labels, and the generator cannot abstain.

7 Conclusion

Large typed decision models are competitive pointwise passage rerankers: Jev is on par with Qwen3-Reranker-8B and ahead of bge-reranker-v2-m3 on average, with large variation across collections (on SPLADE++ candidates it trails on DL19 under standard scoring, a deficit that cannot be separated from incomplete judgments), and its gains over bge persist at equal passage budgets on News and Robust04, as does its Robust04 lead over Qwen3-Reranker-8B when both read the same 4,000 characters (Qwen3-Reranker-8B stays ahead on Touche and News). Same-base controls suggest that Clef's effectiveness comes largely from its underlying LLM: it does not differ significantly from its base on six of seven collections and is ahead only on Robust04,

where the base reads less text, while the small Jeff model improves on its base; Jev's base model is unknown, so we cannot attribute its effectiveness either way. Out of the box, the yes/no probabilities of Jev and Laya are better calibrated than raw cross-encoder scores (Jeff's are not better than bge's, and Jev's graded formats depend on the readout), but a cross-encoder calibrated on other collections with the same relevance threshold reaches similar error; their practical advantage is that no labelled source data are needed, and rewording the prompt moves raw calibration error by more than the gap to bge. Small open decision models remain weak deep rerankers under standard evaluation. For RAG, Jev improves multi-hop answer quality over bge with all three generators, and also over Qwen3-Reranker-8B despite equal ranking effectiveness, and it more often ranks the answer-bearing final hop first; on a family-disjoint MuSiQue split the gains remain (+2.8 to +16.9 F1 over bge, 23 of 27 comparisons significant with family-clustered tests; over Qwen8B, 12 of 27), largest when few passages are selected. Thresholding its probabilities, however, does not improve over fixed top-k selection. Calibration alone, in short, does not make context selection a simple threshold. Future work should build family-disjoint multi-hop splits with more 3- and 4-hop questions, extend the downstream comparison to the API label-probability baselines, and match context budgets across models on all collections.

Supplementary information. Online Resource 1 contains Supplementary Tables S1–S13, Supplementary Fig. S1, and the supplementary analyses cited in the text.

Declarations

Funding. The authors received no financial support for the research, authorship, and/or publication of this article. All API costs were paid personally by the first author; no company, including the providers of the evaluated models and OpenRouter, provided credits, payment, or other support.

Competing interests. The authors have no competing interests to declare that are relevant to the content of this article. None of the authors has a relationship with TypeSafe (Jev), Cloudflare (Clef), firelex (Jeff), ConvAI Innovations (Laya), or OpenRouter.

Ethics approval and consent to participate. Not applicable. The study uses only publicly available benchmark collections and their published relevance judgments, and it involved no human participants, personal data, or animals.

Consent for publication. Not applicable.

Data availability. All collections are publicly distributed and are downloaded automatically by the released code; the underlying Robust04, TREC News, and Signal-1M corpora remain subject to their providers' data agreements. Split IDs, input hashes, and aggregate results are available at https://github.com/parv-bhargava/ChoiceRank.

Materials availability. Not applicable.

Code availability. Code and prompts are available at https://github.com/parv-bhargava/ChoiceRank.

Author contribution. Parv Bhargava: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. Aihan Liu: Validation, Writing – review and editing. Amir Jafari: Supervision, Methodology, Validation, Writing – review and editing.

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